

1 Complete Training Run Registry

1.1 Tool-Use Experiments (Tinker Cloud GPU, Qwen3-8B)

Table 1: Complete tool-use GRPO runs on Tinker (Qwen3-8B, LoRA rank 32). All runs use Adam optimizer ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=1e-8$), max sequence 1024 input / 512 generation.

Run ID	Config	Group	Temp	Steps	Last-10 Reward	Status
88ed2271	Baseline: LR=3e-5	8	0.8	30	0.875	Done
2d488a85	High LR: LR=1e-4	16	0.8	30	0.999	Done
5569c1fa	Low Temp: LR=3e-5	4	0.4	30	0.977	Done
22e9e5fd	xlam-60k: LR=3e-5	8	0.8	30	0.363	Done
386273f4	Synthetic 5-tool	4	0.8	100	0.825	Done
f63b618c	xlam-60k extended	4	0.8	100	0.113	Done
1b01abef	MATH problems	4	0.8	100	0.264	Done

1.2 GSM8K Mathematical Reasoning Experiments (Tinker, Qwen3-8B)

1.3 10x Structural Ceiling Experiments (Tinker, 32 runs, \$65)

1.4 Google Colab QLoRA Experiments

1.5 Advanced RL Algorithm Variants (Colab Notebook)

We provide a reproducible Colab notebook (`advanced_rl_colab.ipynb`) implementing three non-standard RL algorithms for comparison with standard GRPO:

Dr. GRPO [DeepSeek-AI, 2025] removes the KL divergence penalty and adds a token-level entropy bonus ($\lambda = 0.01$) to prevent the entropy collapse documented in Section 5.3 of the main paper. **DAPO** [Yu et al., 2025] further decouples alignment by using asymmetric advantage clipping and higher sampling temperature for diverse rollouts.

DPO [Rafailov et al., 2024] uses rejection-sampled preference pairs (correct vs. incorrect completions from the base model) for offline preference learning without on-policy rollouts.

2 Claim-to-Configuration Mapping

Table 6 maps each main-paper claim to its exact training configuration, enabling reviewers to trace results without cross-referencing run IDs.

Table 2: Complete GSM8K GRPO runs on Tinker. Multi-seed replication (Qwen3-8B), LoRA rank ablation, group size ablation, 4B multi-seed replication, and extended run. All metrics are training-set reward.

Run ID	Model	Seed	Rank	G	LR	Steps	Peak	Last-10
<i>Multi-seed replication (Qwen3-8B, rank 32, $G=4$)</i>								
899d909e	Qwen3-8B	42	32	4	3e-5	50	62.5%	27.5%
5db4e965	Qwen3-8B	137	32	4	3e-5	50	62.5%	27.5%
aabb48cb	Qwen3-8B	256	32	4	3e-5	50	62.5%	32.5%
99971b26	Qwen3-8B	512	32	4	3e-5	50	87.5%	30.0%
b3ba8df6	Qwen3-8B	999	32	4	3e-5	50	87.5%	35.0%
<i>LoRA rank ablation (Qwen3-8B, seed 42, $G=4$)</i>								
ba2a1694	Qwen3-8B	42	8	4	3e-5	50	62.5%	21.2%
92ebcc48	Qwen3-8B	42	16	4	3e-5	50	75.0%	18.8%
9219771c	Qwen3-8B	42	64	4	3e-5	50	87.5%	25.0%
<i>Group size ablation (Qwen3-8B, seed 42, rank 32)</i>								
a13b85f7	Qwen3-8B	42	32	4	3e-5	50	62.5%	23.8%
8c42a6aa	Qwen3-8B	42	32	8	3e-5	50	68.8%	24.4%
ae5c6287	Qwen3-8B	42	32	16	3e-5	50	75.0%	36.2%
6e2f0df3	Qwen3-8B	42	32	32	3e-5	50	100%	54.7%
<i>4B multi-seed replication (Qwen3.5-4B, rank 32, $G=4$)</i>								
b6e8e70d	Qwen3.5-4B	42	32	4	3e-5	50	87.5%	68.8%
566747c0	Qwen3.5-4B	137	32	4	3e-5	50	100%	82.5%
a00e78da	Qwen3.5-4B	256	32	4	3e-5	50	100%	96.2%
cd17f565	Qwen3.5-4B	512	32	4	3e-5	50	100%	91.2%
<i>3B group-size control (Llama-3.2-3B, rank 32)</i>								
86162fb1	Llama-3.2-3B	42	32	32	3e-5	50	12.5%	5.0%
<i>Extended run</i>								
1cc20cec	Qwen3-8B	42	32	4	5e-6	100	75.0%	27.5%

Table 3: Complete 10x Structural Ceiling run registry. All runs are 50 steps on Tinker with full checkpoint resumption. ZVF = Zero-Variance Fraction, GU = Gradient Utilization = 1 - ZVF.

Dimension	Config	Model	Steps	Final Reward	Avg Last-10	Mean ZVF
<i>Benchmark hierarchy (Qwen3-8B, G=16)</i>						
Benchmark	Tool-use JSON	Qwen3-8B	50	1.000	1.000	–
Benchmark	GSM8K (LR=1e-4)	Qwen3-8B	50	1.000	1.000	~1.00
Benchmark	GSM8K (seed4)	Qwen3-8B	50	0.984	0.972	0.550
Benchmark	GSM8K (seed5)	Qwen3-8B	50	0.922	0.925	0.430
Benchmark	MATH-500	Qwen3-8B	50	0.720	0.574	–
Benchmark	HumanEval	Qwen3-8B	50	0.000	0.024	–
<i>Cross-family architecture (8B, tool-use + GSM8K)</i>						
Architecture	Tool-use	Qwen3-8B	50	1.000	1.000	–
Architecture	Tool-use	Llama-3.1-8B-Inst	50	0.103	–	0.00
Architecture	Tool-use	Llama-3.1-8B base	50	0.000	–	1.00
Architecture	GSM8K	Llama-3.1-8B-Inst	39	0.969	–	0.50
Architecture	GSM8K	Llama-3.1-8B base	26	0.047	–	0.75
<i>Model size ladder (GSM8K)</i>						
Size	8B	Qwen3-8B	50	1.000	0.972	0.550
Size	1.7B	Qwen3-1.7B	50	0.016	0.009	0.885
Size	0.6B	Qwen3-0.6B	50	0.016	0.009	0.920
Size	3B	Llama-3.2-3B	47	0.016	~0.02	~0.90
Size	1B	Llama-3.2-1B	50	0.000	0.000	1.00
<i>Group size ablation (Qwen3-8B, GSM8K)</i>						
Group	G=4	Qwen3-8B	50	1.000	0.944	0.520
Group	G=16 (seed4)	Qwen3-8B	50	0.984	0.972	0.550
Group	G=16 (seed5)	Qwen3-8B	50	0.922	0.925	0.430
Group	G=32	Qwen3-8B	50	1.000	0.957	0.455
Group	G=64	Qwen3-8B	50	1.000	~0.98	0.525
<i>Learning rate ablation (Qwen3-8B, G=16, GSM8K)</i>						
LR	1e-5	Qwen3-8B	50	0.594	0.677	0.175
LR	4e-5 (seed4)	Qwen3-8B	50	0.984	0.972	0.550
LR	4e-5 (seed5)	Qwen3-8B	50	0.922	0.925	0.430
LR	1e-4	Qwen3-8B	50	1.000	1.000	~1.00
LR	3e-4	Qwen3-8B	50	0.984	0.901	0.565
<i>Constrained decoding (Qwen3-8B, tool-use)</i>						
Constrained	Unconstrained	Qwen3-8B	50	0.998	–	0.725
Constrained	Constrained	Qwen3-8B	50	0.981	–	0.660

Table 4: Local QLoRA experiments on Google Colab (T4 GPU, 16GB VRAM).

Model	Dataset	Method	LoRA Rank	Steps	Result
Qwen2.5-0.5B	14 synthetic	SFT only	16	10	JSON 30%
Qwen2.5-1.5B	Glaive-v2	SFT+GRPO	16	700	JSON 92%
Qwen2.5-3B	ToolBench-v1	SFT+GRPO	8	40	0.91 multi-turn
Llama-3.2-3B	GSM8K	GRPO only	32	50	2.34% acc
Qwen3-4B	GSM8K	SFT+GRPO	32	50	In progress

Table 5: Advanced RL algorithm configurations (Qwen3-8B + LoRA rank 32 on GSM8K).

Algorithm	Key Change	β (KL)	LR	Temp	Steps
Standard GRPO	Baseline	0.04	4e-5	1.0	50
Dr. GRPO	No KL + entropy bonus	0.0	5e-5	1.0	50
DAPO	Overlong filter + high temp	0.0	4e-5	1.2	50
DPO	Offline preference pairs	0.1	5e-6	–	3 epochs

Table 6: Configuration for each main-paper claim. N = number of independent runs or seeds.

Claim	Model	Rank	G	LR	Steps	N
<i>Original experiments</i>						
Tool 0%→92%	Qwen2.5-1.5B	16	8	3e-5	700	1
Multi-turn 0.72→0.91	Qwen2.5-3B	8	4	3e-5	40	1
Code 32%→40%	Qwen3-8B	32	4	3e-5	50	1
GSM8K 30.5% (SD 3.3)	Qwen3-8B	32	4	3e-5	50	5
Held-out 83.3%	Qwen3-8B	32	4	3e-5	50	5
Group-size scaling	Qwen3-8B	32	4–32	3e-5	50	4
Capacity threshold	3B/4B	32	4,32	3e-5	50	2/4
MoE volatility	Qwen3-30B	32	8	3e-5	50	1
LoRA rank tradeoff	Qwen3-8B	8–64	4	3e-5	50	3
<i>10x Structural Ceiling (32 runs)</i>						
Benchmark hierarchy	Qwen3-8B	32	16	4e-5	50	6
Architecture depend.	Qwen/Llama-8B	32	16	4e-5	50	5
Size ladder	0.6B–8B	32	16	4e-5	50	5
Group saturation	Qwen3-8B	32	4–64	4e-5	50	5
LR tradeoff	Qwen3-8B	32	16	1e-5–3e-4	50	5
Constrained decoding	Qwen3-8B	32	16	4e-5	50	2
SFT prerequisite	Llama-8B base/inst	32	16	4e-5	50	2
Reward hacking	Llama-8B base	32	16	4e-5	50	1

3 Claim-to-Run Mapping

Table 7 maps each main-paper finding to the specific runs that support it, enabling reviewers to trace every claim to its evidence.

Table 7: Mapping of main-paper claims to supporting evidence. Single-run claims are marked (★).

Claim	Runs	Evidence
<i>Original experiments</i>		
Tool calling 0%→92%	Colab Qwen2.5-1.5B	Single SFT+GRPO pipeline (★)
Multi-turn 0.72→0.91	Colab Qwen2.5-3B	4 scenarios, custom rubric (★)
Code 32%→40%	Tinker Qwen3-8B	50-item subset, $p=0.53$ (★)
GSM8K 30.5% (SD 3.3%)	899d, 5db4, aabb, 9997, b3ba	5 seeds, same config
Held-out 83.3% (SD 2.2%)	Same 5 seeds	200 test examples each
Capacity threshold 3B→4B	Colab 3B + 8616 + 4×4B	3B confirmed at both G ; 4 4B seeds
MoE volatility 2.43×	MoE run + 5 dense seeds	Levene’s $p=7\times 10^{-6}$
SFT+GRPO complementarity	Colab 1.5B, 3B	2 pipelines (★)
Group-size scaling	a13b, 8c42, ae5c, 6e2f	4 group sizes, seed 42
LoRA rank–speed tradeoff	ba2a, 92eb, 9219	3 ranks, seed 42 (★)
Two-phase learning	Arithmetic diagnostic run	Single run (★)
Synthetic–real gap	386273f4, f63b618c	2 runs, same model
<i>10x Structural Ceiling (32 runs, W&B: tinker-structural-ceiling)</i>		
Benchmark hierarchy	6 runs across 4 benchmarks	Tool/GSM8K/MATH/HumanEval
Architecture dependence	5 runs (Qwen/Llama-8B)	Tool-use + GSM8K, both families
SFT prerequisite	2 runs (base vs instruct)	Llama-8B, +0.922 delta
Size ladder	5 runs (0.6B–8B)	Qwen + Llama, cross-family
Group saturation (ZVF/GU)	5 runs ($G=4/16/32/64$)	Novel diagnostic metric
LR speed–saturation	5 runs (1e-5 to 3e-4)	Full 50-step correction
Constrained decoding	2 runs	No difference (0.998 vs 0.981)
Reward hacking/collapse	1 run (Llama-8B base)	Loss −238, collapse at step 41 (★)

Key limitation: Of the original twelve claims, four rest on single runs or single pipelines (marked ★). The 10x Structural Ceiling experiment adds 32 systematic runs across 7 dimensions, substantially strengthening the evidence base. The capacity threshold is now confirmed across both Qwen (0.6B, 1.7B) and Llama (1B, 3B) families, partially resolving the architecture confound. The reward hacking observation remains a single run.

Table 8: Tinker SDK hyperparameters for all GRPO runs.

Parameter	Value
SDK Version	v0.16.1
Optimizer	Adam
β_1	0.9
β_2	0.95
ϵ	1e-8
Weight Decay	0
Batch Size	2 prompts/step
Group Size	4, 8, 16, or 32
Max Input Length	1024 tokens
Max Generation Length	512 tokens
Warmup	None (constant LR)
Checkpoint Frequency	Final only

Table 9: LoRA adapter targets and ranks used across experiments.

Parameter	Value
Target Modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
LoRA Ranks Tested	8, 16, 32, 64
Default Rank	32
Alpha	$2 \times \text{rank}$ (default)
Dropout	0.0
QLoRA (Colab)	4-bit NF4, bfloat16 compute

4 Training Hyperparameters

4.1 Tinker Cloud GPU Configuration

4.2 LoRA Configuration

5 Failure Case Analysis

5.1 Three-Billion Parameter Failure Mode

The Llama-3.2-3B model on GSM8K achieved only 2.34% accuracy after 50 GRPO steps (95% CI [0.3%, 4.4%] via binomial proportion on $G=4$ groups), barely above the 1.56% random baseline. A one-sample binomial test yields $p=0.68$, failing to reject the null hypothesis that the model performs at chance. This constitutes a **formal negative result**: GRPO at $G=4$ does not produce statistically detectable learning in 3B models on GSM8K.

The root cause is that **56% of training steps had zero gradient** because all $G=4$ sampled completions received identical rewards (all incorrect). When all completions are equally wrong, GRPO advantages are zero and no gradient update occurs. The 8B model (Qwen3-8B) shows only 16–28% zero-loss rate, enabling effective gradient updates.

Practitioner guideline: Sub-3B dense models should not be expected to learn via GRPO with $G \leq 4$ on math reasoning tasks. Higher group sizes ($G=16$ –64) or curriculum warm-starting may be needed to generate within-group reward variance at this scale, but this remains untested.

Table 10: Zero-loss step percentage across model sizes (lower is better).

Model	Parameters	Zero-Loss %	Final Acc
Llama-3.2-3B	3B	56%	2.34%
Qwen3-8B (seed 512)	8B	16%	30.0%
Qwen3-8B (seed 137)	8B	28%	27.5%
Qwen3-8B (seed 256)	8B	24%	32.5%

5.2 Comprehensive Zero-Loss Dynamics Across All GSM8K Runs

Table 11 reports zero-loss and zero-reward step frequencies across all available GSM8K training runs, providing the quantitative basis for the capacity-threshold and exploration-sparsity claims in the main paper.

Key patterns:

- The 3B model’s 56% zero-loss rate stems from zero-reward (all-incorrect) groups, confirming reward sparsity as the proximate cause.
- The 4B model’s 68% zero-loss rate stems from *all-correct* groups (0% zero-reward), indicating the opposite regime: the model solves problems so easily that

Table 11: Zero-loss and zero-reward dynamics across all GSM8K GRPO runs. Zero-loss indicates all-same-reward groups (no gradient signal). Zero-reward indicates all-incorrect completions. 3B data from Colab; 4B and 8B from Tinker.

Model	Config	Steps	Zero-Loss %	Zero-Reward %	Peak Acc	Last-10 Acc
Llama-3.2-3B	seed=42, G=4	50	56%	56%	3.1%	2.3%
Llama-3.2-3B	seed=42, G=32	50	18%	18%	12.5%	5.0%
Qwen3.5-4B	seed=42, G=4	50	–	–	87.5%	68.8%
Qwen3.5-4B	seed=137, G=4	50	68%	0%	100%	82.5%
Qwen3.5-4B	seed=256, G=4	50	–	–	100%	96.2%
Qwen3.5-4B	seed=512, G=4	50	–	–	100%	91.2%
Qwen3-8B	seed=137, r=32, G=4	50	28%	24%	62.5%	27.5%
Qwen3-8B	seed=256, r=32, G=4	50	24%	24%	62.5%	32.5%
Qwen3-8B	seed=512, r=32, G=4	50	16%	10%	87.5%	30.0%
Qwen3-8B	seed=42, r=32, G=4	50	18%	14%	62.5%	27.5%
Qwen3-8B	seed=999, r=32, G=4	50	18%	16%	87.5%	35.0%
Qwen3-8B	seed=42, r=8, G=4	50	20%	14%	62.5%	21.2%
Qwen3-8B	seed=42, r=16, G=4	50	20%	18%	75.0%	18.8%
Qwen3-8B	seed=42, r=64, G=4	50	18%	16%	87.5%	25.0%
<i>Group size ablation (Qwen3-8B, seed 42, rank 32)</i>						
Qwen3-8B	seed=42, G=4	50	26%	20%	62.5%	23.8%
Qwen3-8B	seed=42, G=8	50	6%	6%	68.8%	24.4%
Qwen3-8B	seed=42, G=16	50	2%	2%	75.0%	36.2%
Qwen3-8B	seed=42, G=32	50	2%	0%	100%	54.7%
Qwen3-8B	lr=5e-6, G=4, 100 steps	100	17%	15%	75.0%	27.5%

group variance vanishes. This is productive zero-loss (learning saturated, not stalled).

- 8B models show 16–28% zero-loss, with seed 512 showing the lowest rate and highest peak accuracy, suggesting exploration varies meaningfully across seeds.
- LoRA rank does not substantially affect zero-loss rates (18–20% for $r=8/16/64$), supporting the paper’s finding that rank affects learning speed but not the fundamental capacity to generate reward signal.
- **Group size dramatically reduces zero-loss:** At $G=4$, 26% of steps have zero gradient; at $G=32$, only 2%. Larger groups ensure at least one correct and one incorrect completion per group, providing consistent gradient signal. This explains why $G=32$ achieves 54.7% vs. $G=4$ ’s 23.8%.

5.3 Qwen3-30B-MoE Volatility Analysis

The Qwen3-30B-MoE model reached 99% peak GSM8K training-step accuracy in our internal runs but exhibited $2.43\times$ higher step-to-step reward variance than the dense 8B model. Statistical testing (Levene’s test) confirms this difference is significant: $F(1, 98) = 22.4, p = 7.0 \times 10^{-6}$.

This volatility stems from MoE routing: different experts activate for different tokens, creating non-smooth loss surfaces. The sparse activation pattern means small changes in the router can dramatically shift which experts process each token, amplifying training instability.

Table 12: MoE vs. Dense training stability comparison.

Model	Step Var (SD)	Peak Acc	Final Acc
Qwen3-8B Dense	0.12	87.5%	30.0%
Qwen3-30B-MoE	0.29	99.0%	95.0%
Ratio (MoE/Dense)	$2.43\times$	—	—

5.4 Policy Entropy, KL, and Group-Composition Diagnostics

The arithmetic task provides the richest telemetry logged during GRPO training. Table 13 summarizes key diagnostic metrics at milestone training steps. Policy entropy collapses by >3 orders of magnitude ($0.773 \rightarrow 0.0002$) within 12 steps, while KL to the training policy remains near zero throughout, indicating the model converges to a near-deterministic policy without drifting far from initialization. Group composition shifts from 81% mixed groups at step 0 (productive gradient signal) to $>96\%$ all-good groups by step 4 (saturated learning, diminishing returns).

Key observations:

Table 13: Diagnostic telemetry from arithmetic GRPO training (Qwen3-8B, 100 steps, G=4). Entropy, KL, and group composition are logged per step; selected milestones shown.

Step	Reward	Entropy	KL (v1)	Frac Mixed	Frac All Good	Frac All Bad
0	0.676	0.773	5.7e-4	0.81	0.19	0.00
1	0.795	0.586	6.3e-4	0.62	0.37	0.01
2	0.933	0.228	-1.4e-4	0.36	0.64	0.00
4	0.999	0.030	9.7e-5	0.04	0.96	0.00
7	0.990	0.004	3.0e-5	0.00	0.99	0.01
11	1.000	2.5e-4	-6.6e-6	0.00	1.00	0.00
19	0.998	0.006	-4.5e-5	0.01	0.99	0.00

- **Entropy collapse:** $0.773 \rightarrow 0.0002$ ($3,865\times$ reduction) in ~ 12 steps. Without explicit entropy regularization, the policy becomes near-deterministic, which limits further exploration.
- **KL remains near zero:** KL divergence to the initial training policy oscillates around ± 0.001 nats, indicating no KL anchor was needed for this short horizon / easy task. Longer or harder tasks may require explicit KL regularization.
- **Group saturation:** Once `frac_all_good` exceeds 0.95, within-group variance drops, advantages shrink, and effective learning rate diminishes (the “zero-advantage” regime). This provides a natural stopping criterion.
- **Missing for GSM8K/MATH:** Comparable entropy/KL/group-composition metrics were *not logged* for the main GSM8K and MATH experiments (Tinker SDK did not expose these in the training loop). Adding this telemetry is a Phase 1 priority in our experimental roadmap (Section 7 of the main paper).

5.5 MBPP Evaluation Failure

The MBPP (Mostly Basic Python Problems) benchmark produced 0% pass rate across all runs due to an output parsing bug in the evaluation script. The script expected a specific JSON format for extracted answers but the model output format differed, preventing correct answer matching.

This represents a methodological failure: an invalidated benchmark should not be reported. Only HumanEval results (which used a working evaluator) are reported in the main paper. Future work should verify evaluator correctness before running experiments.

6 Additional Experimental Details

6.1 Reward Function Weighting Rationale

The tool calling reward ($0.3 + 0.4 + 0.3$) was designed to prioritize:

- **Correct tool name (+0.4):** Selecting the right tool is the core judgment decision
- **Valid JSON (+0.3):** Required for downstream parsing
- **Arguments present (+0.3):** Functional tool use requires populated arguments

These weights were not validated via ablation. Future work should test alternative weightings, particularly increasing the tool-name weight to further emphasize selection judgment over format compliance.

6.2 Evaluation Protocol Limitations

As noted in the main paper, our primary evaluation metrics are computed on **training prompts with new rollouts**, not on held-out test sets. Specifically:

- Tool-use rewards: Fresh sampling from the same prompt distribution
- GSM8K accuracy: Random subset of 8 training examples per step (batch=2, group=4)
- HumanEval: Standard pass@1 on the 50-example benchmark (test set)

The GSM8K training-step evaluation is the most compromised: evaluating on training data with small group sizes ($G=4$) means the “test” examples overlap with training distribution. We subsequently performed held-out evaluation on 200 randomly sampled GSM8K test-set examples (5 seeds, 200 each), reporting 83.3% (SD 2.2%) accuracy. A base-model control (Qwen3-8B without LoRA) scored 82.0% on the same 200 examples, showing the +1.3pp delta is not significant ($p=0.26$).

6.3 Multi-Turn Tool Calling Reward Breakdown

The multi-turn reward function (5-component) enables finer-grained learning signals:

Table 14: Multi-turn tool calling reward components.

Component	Weight
First turn = tool call	+0.25
Final natural language answer	+0.30
Arguments populated	+0.15
Clean JSON output	+0.10
Repeated tool call (penalty)	-0.30

The -0.30 penalty for repeated tool calls was critical: it eliminated SFT’s looping failure mode entirely (Section 4.1.2 of main paper). Without this penalty, models would repeatedly call the same tool in an infinite loop.

Because these scores come from custom scenario rollouts rather than a standard benchmark, we emphasize that they are internal protocol numbers. We did not measure inter-rater reliability, did not use multiple human judges, and did not adopt proxy-state final-answer evaluation. Future revisions should benchmark the same agents under ToolRM / FC-RewardBench-style outcome evaluation or other standardized multi-turn frameworks.

7 Qualitative Examples

7.1 Successful Tool Call (Post-GRPO)

```
<|im_start|>user
What is the weather in Bangalore?<|im_end|>
<|im_start|>assistant
{
  "tool_calls": [
    {
      "name": "get_weather",
      "arguments": {
        "location": "Bangalore",
        "unit": "celsius"
      }
    }
  ]
}<|im_end|>
```

7.2 Failed Tool Call (Pre-GRPO, Plain Text)

```
<|im_start|>user
What is the weather in Bangalore?<|im_end|>
<|im_start|>assistant
The weather in Bangalore is sunny and warm today.
The temperature is around 28 degrees Celsius.
It should be a nice day to go outside!<|im_end|>
```

The pre-GRPO model generates natural language responses instead of structured tool calls. After GRPO training, the model reliably produces valid JSON with correct tool names and populated arguments.

7.3 Two-Phase Learning Trajectory (Representative)

```
Step 1:  Reward=0.00, Loss=0.85, Output="42" (format: plain text)
Step 5:  Reward=0.05, Loss=0.80, Output="The answer is 42." (format: text with n
```

Step 10: Reward=0.10, Loss=0.72, Output="\boxed{42}" (format: boxed answer)
Step 15: Reward=0.14, Loss=0.65, Output="\boxed{42}" (stable format)
Step 20: Reward=0.40, Loss=0.55, Output="\boxed{42}" (reasoning begins)
Step 25: Reward=0.58, Loss=0.40, Output="\boxed{42}" (reasoning improves)
Step 30: Reward=0.55, Loss=0.38, Output="\boxed{42}" (plateau)

This trajectory illustrates the two-phase pattern: format compliance (steps 1-20) precedes reasoning improvement (steps 20-30).

8 Platform and Compute Costs

Table 15: Compute platform usage and estimated costs.

Platform	GPU	Experiments	Est. Cost
Tinker SDK v0.16.1	Cloud A100	25 original runs	\$40–55
Tinker SDK v0.16.1	Cloud A100	32 runs (10x Structural Ceiling)	~\$65
Google Colab Pro	T4 16GB	5 team members	\$10/person
HuggingFace Hub	N/A	Model hosting	Free

Total Tinker spend: ~\$105–120 across 57 runs. Total estimated project spend: ~\$130, well within typical capstone project budgets.

Table 16: Estimated per-run compute budgets for representative experiments. Token estimates assume batch_size=2, group_size as configured, max_generation=512 tokens. Wall-clock times are from logged telemetry where available; others are estimated from Tinker billing.

Domain	Config	Steps	Group	Est. Tokens	Est. Time
Arithmetic	Qwen3-8B, r=32	100	4	410K	~17 min
GSM8K	Qwen3-8B, r=32	50	8	410K	~25 min
GSM8K	Qwen3-8B, r=32	100	8	820K	~50 min
GSM8K	Qwen3.5-4B, r=32	50	8	410K	~20 min
Tool (synthetic)	Qwen3-8B, r=32	30	8	245K	~15 min
Tool (xlam-60k)	Qwen3-8B, r=32	100	8	820K	~50 min

Limitations of budget reporting: We do not report exact tokens processed per run (the Tinker SDK does not expose this metric). Token estimates above are upper bounds assuming maximum generation length. Wall-clock times exclude checkpoint saving overhead. GPU-hours are not tracked per-run; the aggregate across all 57 Tinker runs is estimated at 24–32 A100 GPU-hours (original 25 runs: 12–16 hours; 10x Structural Ceiling 32 runs: ~12–16 hours at ~40 min wall time for parallel batch).

We report GPU platform and wall-clock budget, but not a full token-processed accounting per run. Likewise, our data-split story is uneven across domains: HumanEval

uses a held-out benchmark subset, while tool calling and GSM8K largely report training-distribution rollouts. This split mismatch is a major threat to cross-domain comparability.

9 Code and Reproducibility

All training scripts are available in the repository:

- `grpo_gsm8k_base.py` – Parameterized GSM8K training
- `grpo_exp_a_baseline.py`, `exp_b_high_lr.py`, `exp_c_low_temp.py`, `exp_d_xlam.py` – Tool-use experiments
- `grpo_100_xlam.py`, `grpo_100_synthetic.py`, `grpo_100_math.py` – Extended runs
- `experiments/10x_structural_ceiling/grpo_10x_runner.py` – Unified runner with YAML config, checkpoint resumption, ZVF/GU logging
- `experiments/10x_structural_ceiling/configs/*.yaml` – 23 YAML configs for all 10x ablation dimensions
- `experiments/10x_structural_ceiling/RESULTS.md` – Complete 50-step results with 8 findings

At present, we can release training scripts, paper sources, and the hardened held-out evaluation script, but the exact prompt packs / scenario definitions used for every internal tool-calling score are not yet packaged as a replication bundle. Releasing prompts, scoring rubrics, and scenario manifests is an explicit next step for reproducibility.

For stability diagnostics, we did not log KL-to-SFT trajectories or policy entropy, and our MoE analysis does not yet include routing entropy or expert-load imbalance measurements. Those omissions limit causal interpretation of both drift and volatility claims.

Representative training logs were collected during experimentation, but the literal local `/tmp` paths used during development are not a portable reproduction interface and should not be treated as a release artifact.

Checkpoint access is partial rather than universal at present:

- Some trained artifacts were published to project-managed model hosting during development.
- Some tool-use and math checkpoints were written to Tinker under run-specific paths such as `tinker://run_id/sampler_weights/final`, but present-day recovery depends on retention and account access.

A clean anonymous release should replace personal account identifiers with project-level artifact locations or explicit recovery instructions.

10 Content Moved from Main Paper

10.1 GRPO Hyperparameter Comparison Table

Table 17: Standard GRPO hyperparameters across published systems.

System	G	LR	T	β (KL)
DeepSeek-R1-Zero	16	3×10^{-6}	1.0	0.001
DeepSeekMath-RL 7B	64	1×10^{-6}	–	0.04
JT-Math	16	4×10^{-6}	1.2	0.001
Skywork-OR1	–	–	–	0
R1-style (Qwen/Llama)	8	$2.5\text{--}5 \times 10^{-6}$	0.7	0
Ours	16	4×10^{-5}	1.0	–

Our $\sim 10\times$ higher learning rate is attributed to LoRA-only training (0.1% trainable parameters vs. full fine-tuning in published systems).

10.2 Reward Function Limitations

Our reward functions are deliberately simple and verifiable, but this coarseness introduces known limitations. The tool-calling reward checks JSON validity, tool name, and argument-key presence but does *not* verify argument-value correctness—a model can receive full reward by calling the right function with plausible but wrong values. This creates a reward-hacking surface. Adding argument-type and value checks, or leveraging outcome reward models such as ToolRM, would reduce this risk. The math reward is binary (correct/incorrect) with no partial credit for intermediate reasoning steps, contributing to reward sparsity for weaker models.

10.3 Safety Considerations

GRPO-trained tool-calling models introduce safety risks: (1) **incorrect tool execution**—valid JSON with wrong functions or dangerous arguments; (2) **persistent loops**—the -0.30 penalty eliminated loops in our experiments but may not suffice for complex workflows; (3) **hallucinated capabilities**—calls to nonexistent tools, especially when transferring from synthetic to real schemas. Future work should incorporate hallucination and loop penalties validated by standardized evaluators.

10.4 Practitioner Decision Guide

Based on our findings, we offer preliminary guidance for practitioners considering GRPO for post-training (Table 18).

Caveat: These guidelines are derived from experiments on Qwen/Llama families with GSM8K and custom tool-calling tasks. Generalization to other model families, harder benchmarks, or multi-turn agentic trajectories is untested.

Table 18: Practitioner decision guide based on empirical findings. All recommendations are preliminary and derived from limited experiments.

Condition	Recommendation	Evidence
Model $\leq 3B$	Do not use GRPO ($G=32$ yields only 5%)	Confirmed at $G=4$ and $G=32$ (Sec. 5.1)
Model 3B–4B	Use $G \geq 16$; consider SFT warm-start	Capacity threshold (Sec. 5.1)
Model $\geq 8B$	GRPO viable; $G \geq 16$ recommended; rank 32 sufficient	5-seed replication + group ablation
MoE architecture	Expect $2\text{--}3\times$ reward variance	$p=7\times 10^{-6}$ (Sec. 7)
Tool calling	SFT first, then GRPO	$0\%\rightarrow 92\%$ (Sec. 4.1)
Synthetic data	Expect $3\text{--}8\times$ easier than real	Sec. 5.2
LoRA rank choice	Rank 32 default; 64 for faster convergence	Rank ablation (Sec. 4.4.3)

10.5 Planned Experiments and Evaluation Roadmap

Phase 1 (Immediate): Held-out GSM8K test evaluation (1,319 examples, 3+ seeds, bootstrapped 95% CIs) and MATH test set transfer.

Phase 2 (Ablations): Group size sweep $G \in \{4, 8, 16, 32\}$ on 8B completed; remaining: $G \in \{64, 128\}$, group-size ablation on 3B, temperature sweep $T \in \{0.7, 1.0, 1.3, 1.5\}$; KL anchoring $\beta \in \{0.01, 0.05, 0.1\}$; curriculum (easy-to-hard).

Phase 3 (Benchmarks): Canonical HumanEval-164/MBPP with pass@ k ; FC-RewardBench-style tool evaluation; SFT-only, PPO, DPO, RLOO baselines under matched compute. Reproducible notebooks (Dr. GRPO, DAPO, DPO, PPO, REINFORCE) are available in the supplementary repository.

Phase 4 (Diagnostics): MoE routing telemetry; LoRA rank $r \in \{8, 16, 32, 64\}$ with memory/throughput/accuracy; QR-Adaptor comparison.

Estimated scope: $\sim 20\text{--}30$ runs, \$500–\$1000.

10.6 Reviewer Priority Analysis

A systematic analysis of 490 RL-for-LLM reviews across 248 papers (2017–2023) finds that *insufficient evaluation* is flagged in 86.3% of reviews, with *missing baselines* (56.8%) and *missing ablations* (14.5%) as the top complaints. Human evaluation absence is rarely criticized (2.1%), and including capability benchmarks shows no significant correlation with reviewer scores (Mann–Whitney U , $p > 0.05$). This motivates our prioritization of baseline comparisons and ablation coverage over human-judge studies.

11 Tool-Calling Evaluation Protocol

This section documents the custom tool-calling evaluation setup used in the main paper, addressing the reviewer concern about non-standardized evaluation.

11.1 Tool Schema Format

All tool schemas follow the OpenAI-compatible function-calling format:

```
{
  "type": "function",
  "function": {
    "name": "get_weather",
    "description": "Get current weather for a location",
    "parameters": {
      "type": "object",
      "properties": {
        "location": {"type": "string", "description": "City name"},
        "unit": {"type": "string", "enum": ["celsius", "fahrenheit"]}
      },
      "required": ["location"]
    }
  }
}
```

11.2 Single-Turn Evaluation Rubric

Each tool call is scored on three binary components:

Table 19: Single-turn tool-calling reward rubric.

Component	Weight	Criterion
JSON Valid	+0.3	Output parses as valid JSON
Correct Tool	+0.4	name field matches expected tool
Arguments Present	+0.3	All required argument keys exist

Limitations: This rubric does *not* verify argument values, argument types, or semantic correctness. A model can score 1.0 by producing syntactically correct JSON with plausible but wrong argument values.

11.3 Multi-Turn Scenario Set

Four hand-authored scenarios with deterministic expected tool sequences:

11.4 Prompt Template

```
<|im_start|>system
You are a helpful assistant with access to the following tools:
{tool_schemas_json}
```

When you need to use a tool, respond with a JSON object

Table 20: Multi-turn evaluation scenarios.

Scenario	Expected Tool Sequence	Turns
Weather + Packing	get_weather → suggest_packing	2
Stock + News	get_stock_price → get_news	2
Search + Calculate	web_search → calculator	2
Single Tool	get_weather	1

containing "tool_calls" array. Otherwise respond normally.
<|im_end|>
<|im_start|>user
{user_query}<|im_end|>

11.5 Judge Details

Scoring is fully automated (no LLM judge, no human annotation):

- JSON validity: Python `json.loads()` parsing
- Tool name: exact string match against expected tool
- Arguments: set intersection of required keys
- Repeat penalty: -0.30 if name matches any previous turn

Inter-rater reliability was not measured because scoring is deterministic. This is a *limitation*: deterministic rubrics can miss semantic correctness that human judges would catch.

References

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